Context-Neutral Linear Weights (wOP+) vs. Environmentally Adjusted Metrics (wRC+)

Research Question: *Why do modern baseball front offices heavily favor weighted metrics influenced by environmental variables (like wRC+) versus raw, context-neutral linear data (Weighted Offensive Production (my stat) (wOP+)? What are the strategic pros and cons of each, and how can a hybrid approach improve talent evaluation and roster construction?*

wRC+ is a widely known and analyzed stat in Major League front offices that is used to help determine a players worth and offensive productiveness. One of its main ingredients that makes it stand out from the rest, is its variable weights that's inside its equation, including ball park factors, and LR/BA (league run average/ plate appearance is roughly 0.11 to 0.12 per PA, which means a team roughly scores 1 run every 8 or 9 plate appearances.). But what would happen, if someone created a statistic that used context-neutral linear weights (like run expectancy per raw stat), and compared that stat to wRC+, what would be the outcome? Well, that's exactly what I set out to do, by creating Weighted Offensive Production (wOP+)!

For the weighting system of my stat, I pulled from FanGraphs "Guts!" section, and used the run expectancy weights for these stats: BB, HBP, 1B, 2B, 3B, HR, SB, and CS. I then averaged players' scores on the same scale as wRC+ uses, making a "100" score the average player's score. I kept the minimum plate appearances to 500, to keep things consistent. I wanted to analyze everyday players for this analysis.

Here is my first pull of data from my github repository showing the top 20 Players for wOP+ from 2026:

	Name	PA	HR	RBI	Offensive_Production_Plus
14	Yordan Alvarez	700	42	106	132
130	Pete Crow-Armstrong	734	45	107	123
654	James Wood	623	30	77	119
479	Ben Rice	677	43	103	119
119	Willson Contreras	543	28	82	118
424	Shohei Ohtani	618	30	81	116
11	Pete Alonso	700	43	113	115
47	Jake Bauers	578	26	86	115
138	Elly De La Cruz	636	30	79	115
24	Randy Arozarena	670	26	79	114
88	Junior Caminero	702	42	101	114
525	Kyle Schwarber	710	46	98	113
230	Bryce Harper	702	30	89	113
146	Rafael Devers	633	37	98	113
401	Gabriel Moreno	549	16	83	112
610	Miguel Vargas	714	33	93	111
184	Ty France	508	21	72	111
404	Max Muncy	561	29	76	111
0	CJ Abrams	654	33	106	110
571	Seiya Suzuki	642	26	86	110

Here is the list of the top 20 Players based off of wRC+ from 2026:

#	Name	Team	G	PA	HR	R	RBI	SB	BB%	K%	ISO	BABIP	AVG	OBP	SLG	wOBA	xwOBA	wRC+
1	Yordan Alvarez	HOU	158	692	42	105	106	1	15.8%	17.3%	.286	.334	.316	.432	.601	.429	.442	181
2	Pete Crow-Armstrong	CHC	162	726	45	119	107	41	11.4%	25.8%	.291	.327	.280	.372	.570	.400	.374	157
3	Randy Arozarena	SEA	154	670	26	113	79	26	10.1%	21.6%	.203	.333	.281	.381	.484	.379	.348	151
4	Willson Contreras	BOS	130	535	27	68	81	3	11.0%	25.2%	.241	.332	.275	.386	.516	.391	.376	151
5	Ben Rice	NYY	154	667	41	105	97	3	12.7%	24.3%	.271	.295	.264	.363	.534	.384	.378	149
6	Pete Alonso	BAL	161	700	43	100	113	5	11.6%	24.1%	.254	.299	.267	.363	.522	.379	.402	148
7	Jake Bauers	MIL	144	578	26	93	86	11	16.3%	27.0%	.218	.339	.266	.389	.484	.381	.371	143
8	Junior Caminero	TBR	162	702	42	100	101	3	11.3%	18.4%	.248	.286	.276	.360	.524	.375	.364	142
9	James Wood	WSN	135	623	30	106	77	21	16.4%	28.7%	.238	.344	.265	.388	.503	.383	.403	142
10	Gabriel Moreno	ARI	130	549	16	72	83	6	11.1%	14.6%	.175	.339	.311	.388	.486	.380	.363	141
11	Ty France	SDP	133	500	21	59	72	1	7.4%	21.6%	.207	.342	.295	.363	.502	.373	.327	141
12	Elly De La Cruz	CIN	143	636	30	99	79	30	11.3%	27.5%	.233	.359	.283	.365	.515	.380	.376	139
13	Shohei Ohtani	LAD	139	618	30	89	81	11	13.1%	24.6%	.237	.329	.275	.377	.512	.377	.397	139
14	Riley Greene	DET	140	592	21	79	75	3	12.0%	26.5%	.194	.361	.280	.370	.474	.366	.369	136
15	Rafael Devers	SFG	148	633	37	88	98	0	10.9%	25.4%	.267	.294	.258	.340	.525	.367	.342	136
16	Bryce Harper	PHI	162	693	29	93	87	7	15.4%	21.4%	.217	.309	.269	.380	.485	.375	.379	135
17	Miguel Vargas	CHW	159	704	33	106	93	22	14.3%	17.9%	.224	.261	.250	.363	.474	.363	.378	134
18	Dominic Canzone	SEA	146	533	23	60	77	2	9.0%	20.6%	.220	.297	.264	.336	.484	.354	.359	134
19	Seiya Suzuki	CHC	143	634	26	96	86	3	12.8%	24.4%	.204	.330	.270	.368	.473	.366	.340	134
20	Chase DeLauter	CLE	133	559	16	62	67	11	10.7%	14.7%	.170	.315	.287	.363	.457	.359	.346	131

I am going to take both of these top 20 players for wOP+ and wRC+ and put them into a Scatter Plot, and find the outliers. I already noticed some players who are in one list, but not the other.

IN wOP+ but not wRC+

- **Kyle Schwarber** (Ranked #12 on your list with 113 wOP+) (he's 22st in wRC+)
- **Max Muncy** (Ranked #18 on your list with 111 wOP+)
- **CJ Abrams** (Ranked #19 on your list with 110 wOP+)

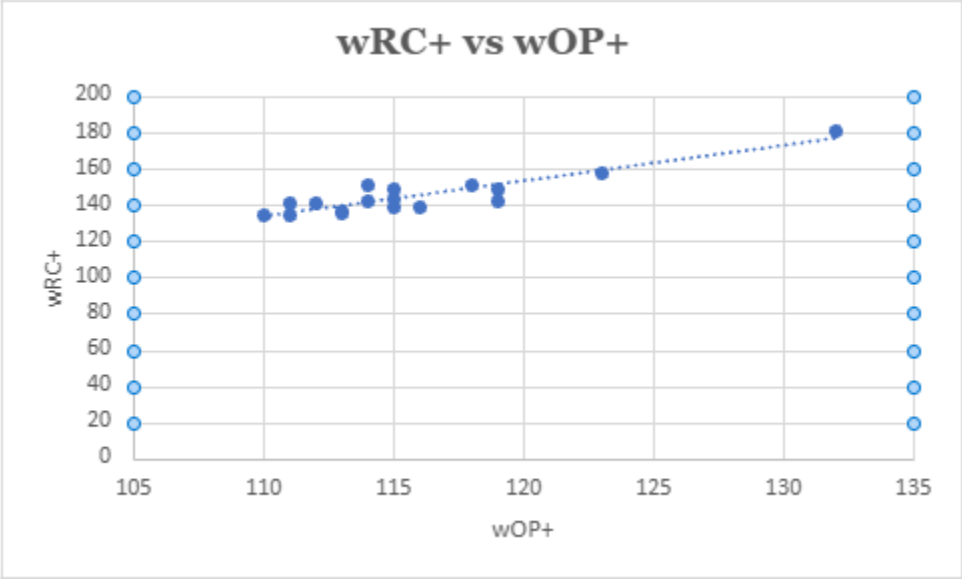
IN wRC+ but not wOP+

- **Riley Greene** (Ranked #14 on FanGraphs with 136 wRC+)
- **Dominic Canzone** (Ranked #18 on FanGraphs with 134 wRC+)
- **Chase DeLauter** (Ranked #20 on FanGraphs with 131 wRC+)

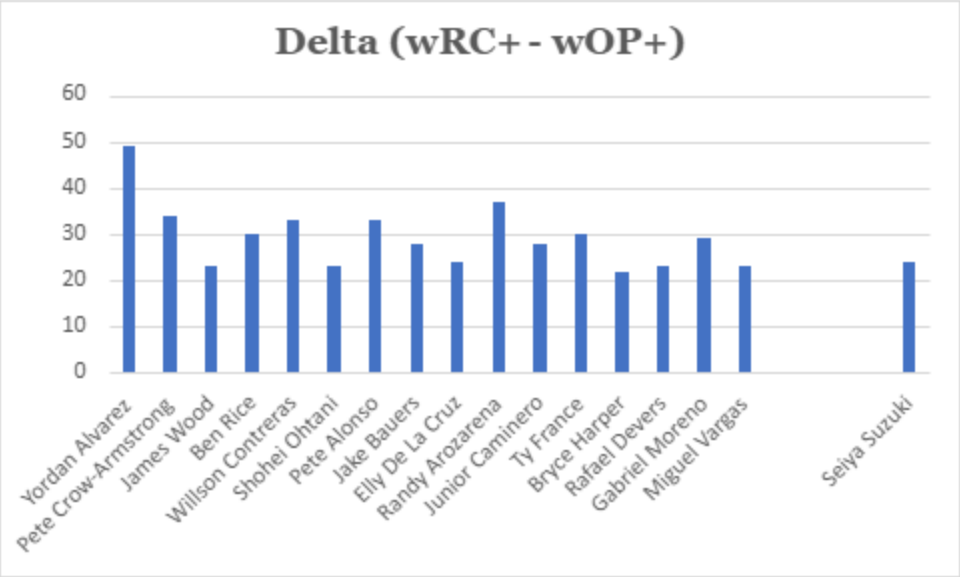
These outliers are people I am going to research the most into, as there are some slight variances with the other players on both lists, in fact, 17 out of the 20 players on both lists match, so there is a 85% alignment between the two scores when looking at Top 20 Players, now it's just time to dig deeper as to why these 6 players are the odd balls (in or out).

My Analysis:

After inputting both top 20 lists into excel, I have come up with these two charts for the 17 matching players in both lists:



This chart above is a correlating scatter plot, on the x axis, my wOP+ stat, and on the Y axis, the widely known wRC+ stat. The players are not named, but it shows the correlation and the relative similarity between the two scoring stats. Now the chart below, shows the Delta between the two:



This chart tells a much different story. The biggest outlier here is Yordan Alvarez. His delta is +49, the biggest out of the 17 players. The rest of them are all in the +20-35 range, but this tells a story. My wOP+ strictly weighs players performances on a linear weight, based solely on walks, singles, doubles, triples, home runs, stolen bases, and caught stealing attempts. The gap between the two, is how wRC+ inputs its environmental factors, how it weighs ball park metrics, whether a hitter is hitting well in a pitcher's park or vice versa, and the League Runs per Plate Appearance for that given season. Those factors show in the biggest of deltas (Yordan Alvarez), and it is a good contrast between the more concise, conservative wOP+, and the more heavily weighted and always changing wRC+.

The Outliers:

Here is where the interesting diversions happen when looking at the outlier players. For each one, I am going to analyze their wOP+ and wRC+ difference, make a hypothesis for the why there's a difference, and I will look at a few determining factors to see why they are distinct from one another, and to see if there is some underlying stats that are getting over looked: The ball park they play in, the League Runs per AB for 2026, and their Statcast numbers (exit velocity, average launch angle, hard hit ball rate, etc.). I will analyze the first 4 players that are in my wOP+ top 20, but are NOT, in the top 20 for wRC+. First up, Kyle Schwarber.

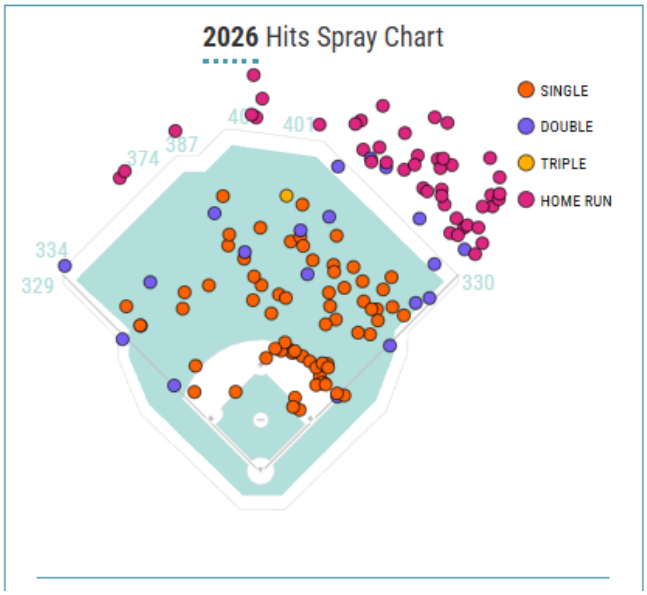
Kyle Schwarber

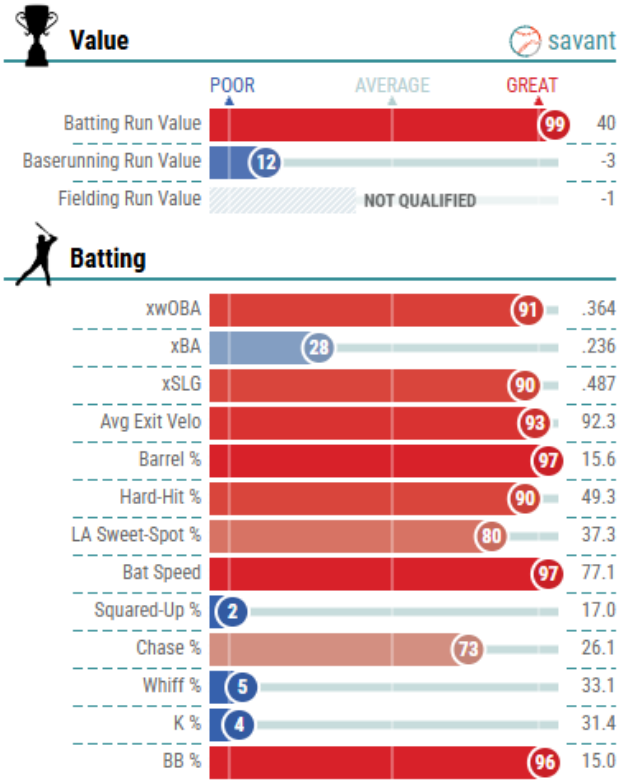


wOP+: 113+
wRC+: 130+

Hypothesis: Kyle plays in Philadelphia, one of the more hitter friendly parks, especially for pull side lefties like himself, he fits perfectly there with his swing. I think the ball park factor is a bigger drawback for him when looking at his wRC+.

Analysis: Kyle barely missed out on the wRC+ top 20, ranking 22nd. I think his wRC+ rating and wOP+ ratings are valid. The reason why he did not break into the top 20 is mainly due to one factor: where his home team plays. He is a huge pull side hitter, and Philadelphia favors lefty pull hitters, as you can see for his spray chart:





It's a decently short porch, and regardless where he plays, he ranks in the 99th Percent Value for Batting Run Value on Baseball Savant, Kyle is property rated in both stats. A good comparison to make for Kyle is Junior Caminero. Junior ranks 12th in wOP+, but ranks 8th in wRC+. The main reason why? Tropicana Field is a much harder place to be a good hitter than Citizens Bank Park is.

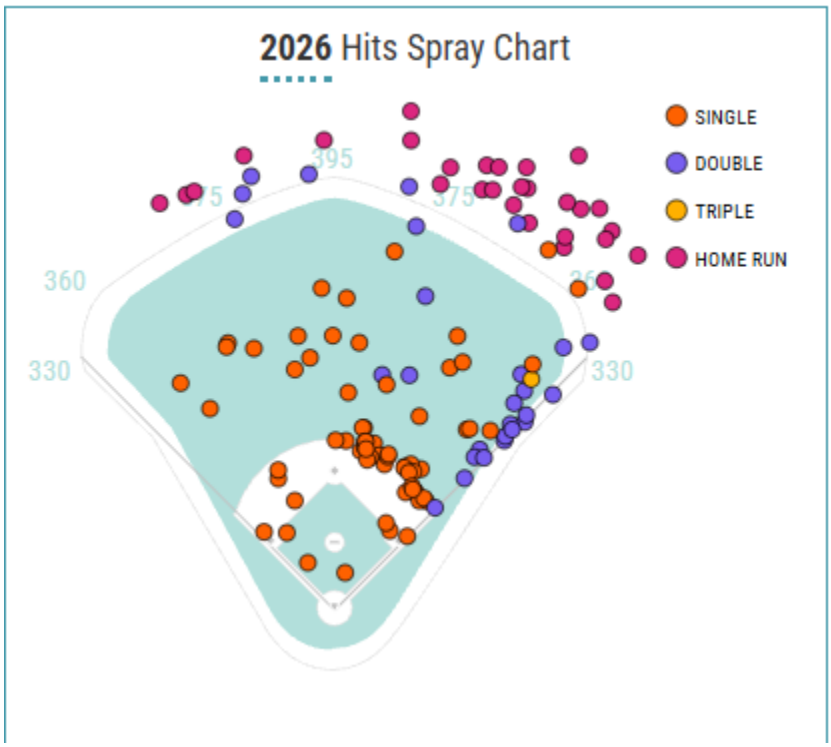
Max Muncy (the anomaly in the data)



wOP+: 111
wRC+: 129

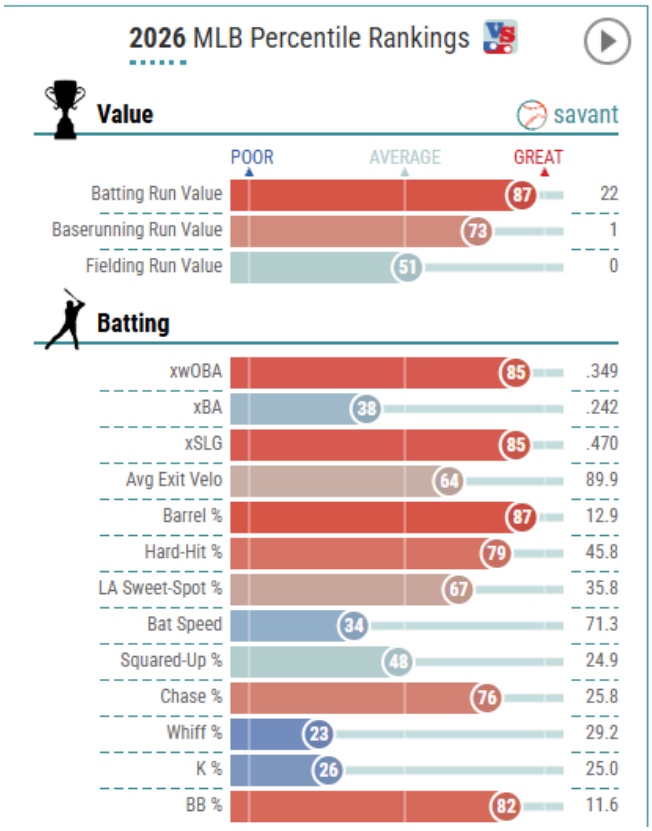
Hypothesis: He is 18th in wOP+, but 28th in wRC+ with a 129+. It's interesting, he is 2% more average than Kyle Schwarber is when looking at wOP+ (2 points less than Kyle), but he is only 1% more average than Kyle when looking at wRC+. What I think is going on here, is that Max plays in a slightly harder hitter environment in LA, which is why wRC+ is giving him a boost in points, but is raw stats are slightly worse than Kyle's, hence the lower wOP+. Let's look deeper:

Analysis: There is definitely something interesting going on with Max's case here when comparing it to Kyle's stats. So yes, Max DOES play in a harder hitting environment in LA, he is a pull hitter, take a look at his spray chart:



That right field fence is on average 25-30ft farther than Citizens Bank Park. But here is something very very interesting. When looking at Max's Baseball Savant, his

metrics do not pop off like Kyle's do:



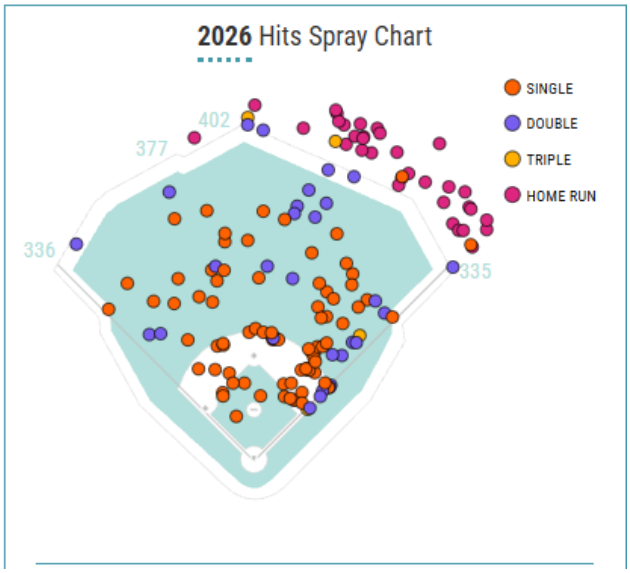
His batting run value only sits at 87, which is still super great. But his avg exit velo is in the 64th percentile, and Kyle's is in the 93rd percentile. Almost every major metric here, like xSLG, Barrel %, Sweet Spot%, etc., are all substantially lower than Kyle Schwarber's, but Kyle only has a 1% buffer in wRC+ than Max does? I think this is a great example of how wRC+ can be a flawed metric when not taking into account other stats like Baseball Savant presents. Again, Kyle plays at a more hitter friendly park, but you can't argue with statcast data, Kyle just swings the bat better than Max does, but is only granted a 1% buffer above him. That being said, using baseball savant also hinders wOP+'s validity too, even though there is a 3% buffer between the players, using savant gives a full picture of who really is the better hitter.

CJ Abrams

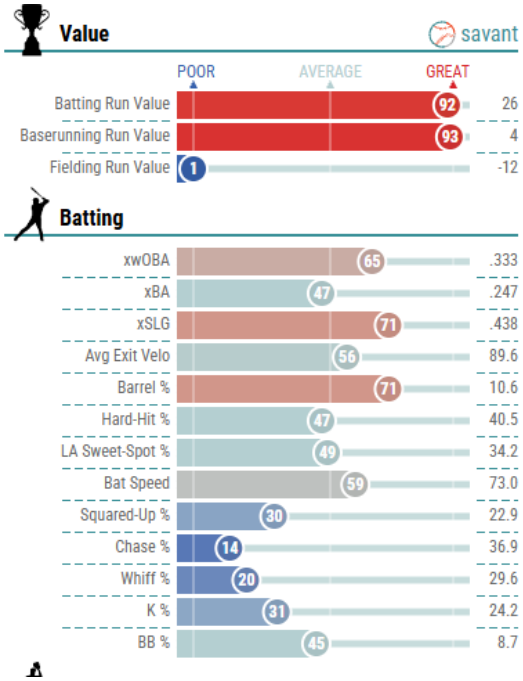
wOP+: 110
wRC+: 124

Hypothesis: CJ plays at a pitcher friendly park, so I believe that is going to be one of the bigger factors into why he is ranked 35th in wRC+, but 19th in wOP+.

Analysis: CJ's speed definitely is what is keeping his score's relatively high, that and his ability to pull home runs in Washington D.C. Here is his spray chart:



Another left handed heavy pull hitter here. CJ is a different player composition than Max and Kyle, he's kind in a league of his own. He's in a left handed friendly ball park, and he is wicked fast, stealing 34 bases this year, and hitting 33 home runs, almost all of them to the pull side. He doesn't particularly hit the ball extremely hard though:



His OBA and SLG and Barrel are above league average, but his other hitting stats are average, or even below average. I think his wOP+ and wRC+ are weighted fairly for him, given his speed and ability to get on base makes him a more dangerous offensive threat.

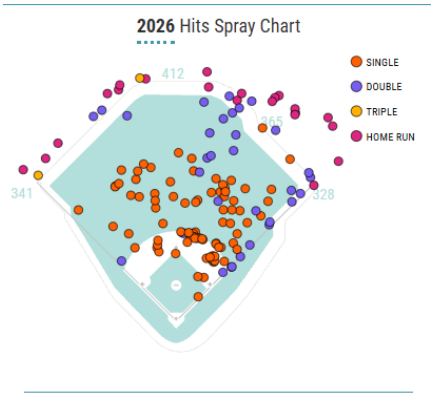
Riley Green



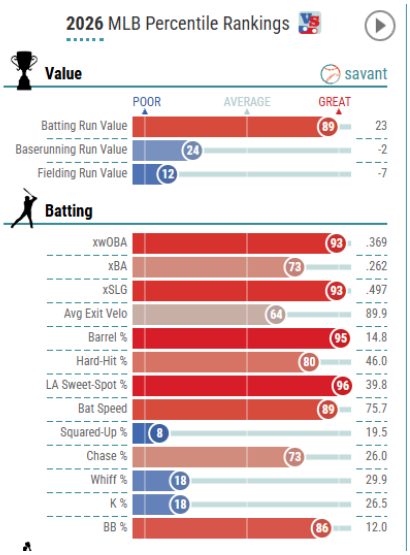
wOP+: 110
wRC+: 136

Hypothesis: Riley barely missed out in the top 20 for wOP+, sitting at 21st, right behind Seiya Suzuki. But he is ranked 14th in wRC+ with 136, big jump for him. He is 6% better than Kyle Schwarber, which makes me think: Comerica park has to have a huge play in his wRC+, super pitcher friendly place. Let's look into the data:

Analysis: As assumed, Riley is a great, great hitter. He always makes good contact, he gets on base a lot, and that's attributed not only to his high number of hits, but he is in the 86th percentile in walks, which is a key stat used in both statistics. He hits the ball around, per his spray chart:



He has only 21 home runs, which could be attributed to the bigger field he plays at. But I also think this is a good example of wRC+ adjusting for park factors. Take a look at his statcast numbers:



Riley has very good bat-to-ball skills, a little less than Kyle does, but much more than Max does on average. And with the adjustment with the park factor, it goes to show that wRC+ for Riley Greene does really show how good of a hitter he really is, and Baseball Savant backs up that claim. And his wOP+ is still really solid, 10% above the average hitter.

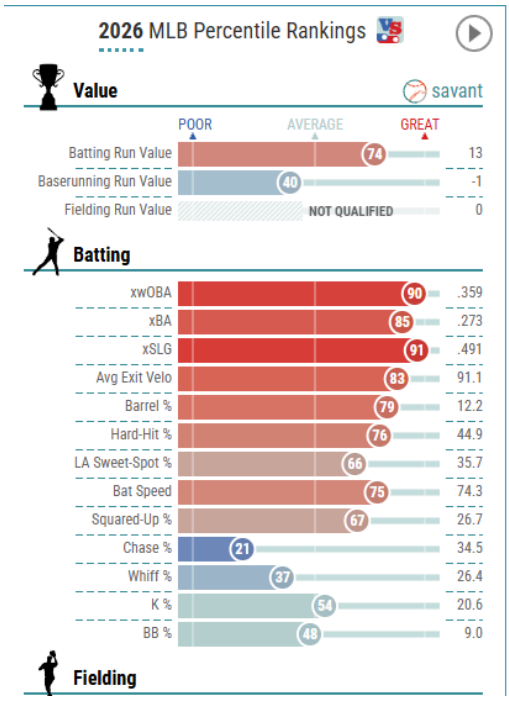
Dominic Canzone



wOP+: 107
wRC+: 134

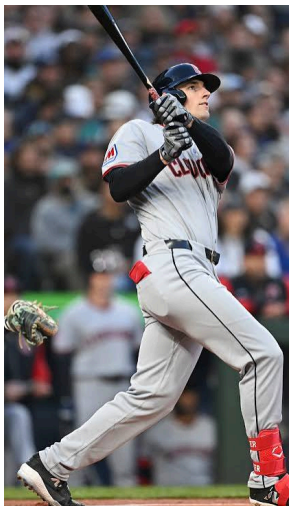
Hypothesis: Since Dominic plays in Seattle – arguably the hardest place in baseball to be a perennial offensive player – I think that has a big factor as to why his wRC+ is as high as it is. Comparing his wOP+ to a player like Kyle Schwarber, he is 6% more average, but Kyle plays in Citizen Bank compared to T-Mobile, pretty drastic hitting difference between the two parks, let's see the analysis:

Analysis: Digging into Dominic's statcast, he is a very solid bat-to-ball player, ranking 80th minimum percentile for a few of the hitting categories. His batting value, however, is lower than the previous players analyzed, and that can be attributed to his patience at the plate:



He's below average in taking walks, and average in K rates, but his whiff and chase rates lower his batting value according to Baseball Savant. Taking these stats into account, and analyzing what wOP+ (34th) and wRC+ (18th) gave him, I think he is still a very solid hitter, and if he played in Cincannatti or in New York, his name would be a lot bigger in the game, as the park would factor into his performance greatly. I will say though, as a personal preference when evaluating players, regardless of how high wRC+ rates him, his chase, whiff, and walk rates do alarm me, especially for mainly being a DH, I think those are habits to look out for when analyzing this player that wRC+ can't account for.

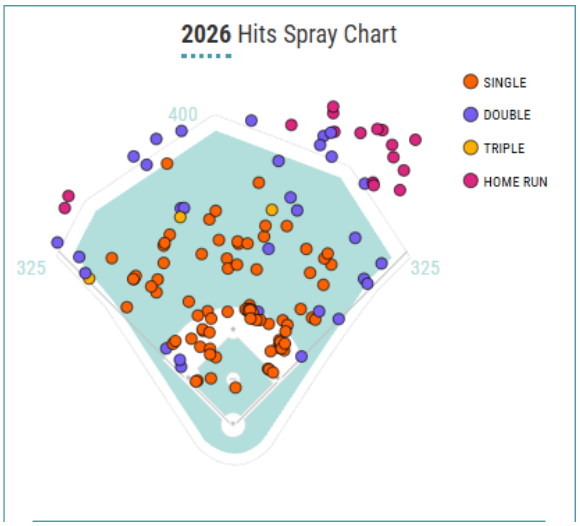
Chase DeLauter



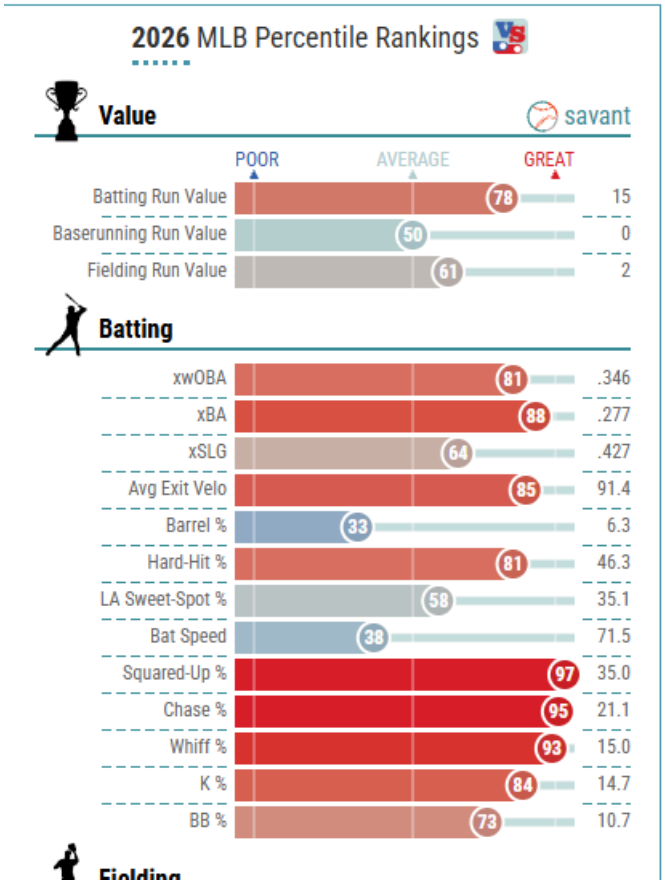
wOP+: 107
wRC+: 131

Hypothesis: Chase scrapped into the top 20 for wRC+, slightly out beating Kyle Schwarber to make it in. Chase does play in a slightly more favored pitcher's park in Cleveland, but not so heavily as Seattle. I think the Cleveland park factor will be the main reason why his wRC+ is high, but his wOP+ is practically the same as Dominic, so I am curious what digging more into him will tell me about these stats he has:

Analysis: So Chase is in his rookie season, so there isn't much data we can pull from. But his scatter chart definitely shows where he liked to hit the majority of his 16 home runs:



As you can see, when he hit a home run, he really got into it. But he hit for 30 doubles this year, 15th in the American League, but the most impressive thing about this player, is his patience, unlike our previous player:



His chase, whiff, K, and BB rates are all outstanding, especially for a rookie, that takes a lot of poise to achieve, which is where I am thinking his wRC+ and his OBP (10th in Baseball) led him to have a slight edge over Kyle, and him playing in a harder hitting environment. But his other stats above are overwhelmingly impressive, but his barrel and bat speed are confusingly low. Overall though, the delta between his wOP+ and his wRC+ do make sense, though I think him slightly being higher than Kyle in wRC+ doesn't make a whole lot of sense, but he should definitely be higher than Max Muncy. I don't have the exact park factor equation and LR/BA data to see exactly why he is 1% better than Kyle, but he is 7% more average in raw stats compared to Kyle, so it's interesting debate to see which player provides more value to a team, depends on the preference of the front office I guess.

Conclusion

Going back to my research question, I think stats like wOP+ that are context-neutral, and just take a players raw stats, can still be a very useful way to assess a player's ability, but it's not without it's drawbacks, like inflating players who play in easy ball parks for example. On the flipside, wRC+ is a heavily used statistic in today's game, and it has its perks, but there are some players who fall through the cracks, like Max Muncy for example. His park factor benefited him greatly, but taking a look at his statcast metrics, and using wRC+ to judge him solely, would be a mistake in my eyes.

I also think it is important to note, that yes, these outlier players had reasons to not be in the other top 20 list, but all of them are elite players. For the most part, if one of these outliers were not in the top 20 of one of the stats, they were definitely in the top 40 of it. What I am saying is, I am analyzing marginal outliers in this data, just for context. But the importance of breaking each individual player down, and creating a profile for them, brought their weighted stats and statcast data to life, and it allows a scout, a front office executive, and managers to build a more sustainable way of analyzing player talent.

Overall, I believe that front offices should not just look at one, or even two stats as their main grading scale when analyzing players (and the good front offices do not do this), but they should look at all players stats, and most importantly, while the data and stats can prove a players productiveness, the human touch and analysis of a player on the field playing the game, is the most crucial key to the puzzle. Because baseball, even though it's very data driven now, at its core, is a human experience, and it is played by, watched by, and loved by humans.